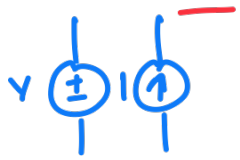
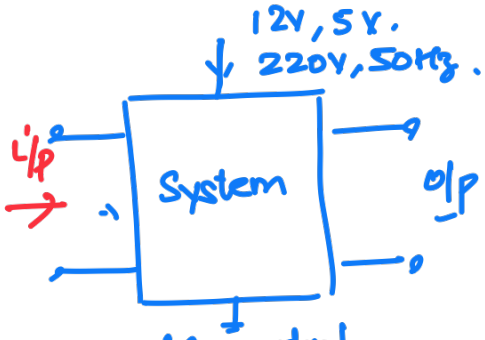
Lecture #02

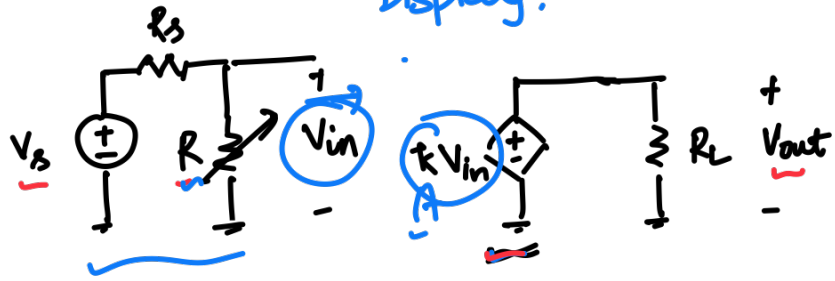


@ $\omega_{in} = 2\pi f_{in}$



- Cooling cap. (Ton)
- Loudness (dB)
- Brightness ( )





- $R: [0, \infty] \Rightarrow V_{in} \in \{0, V_s\} \Rightarrow V_{out} \in \{0, kV_s\}$ (A)

- $P = \left(\frac{V_s}{R_s + R}\right)^2 \times R$ across tunable resistor

- $P_{in} = 0$

- $P_{out} = \frac{(AV_{in})^2}{R_L}$ ✓

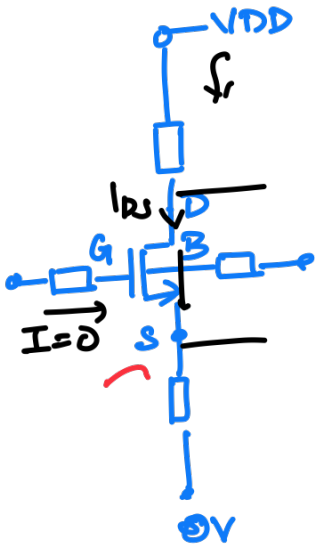


$$kV_{in} = f(V_{dd}, V_{in}) = f_1(V_{dd}) + f_2(V_{in})$$

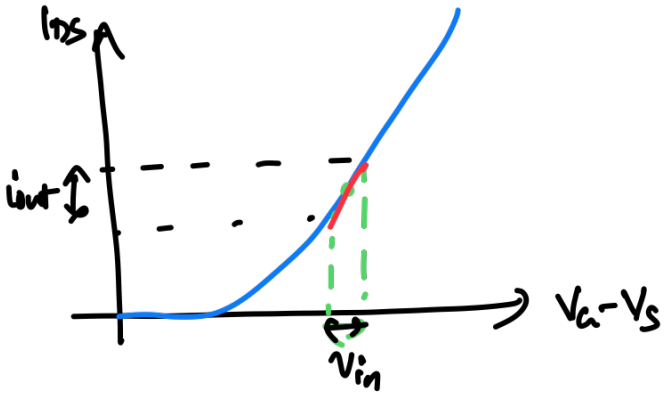
$f_1(V_{dd})$ (1) $f_2(V_{in})$

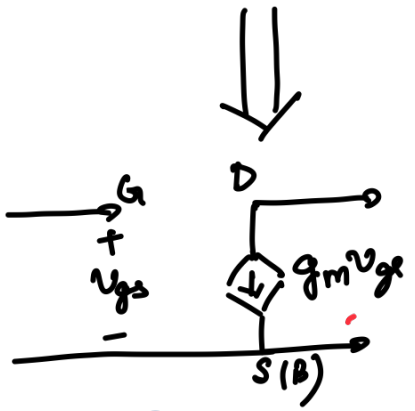
$$= k \times V_{in}$$

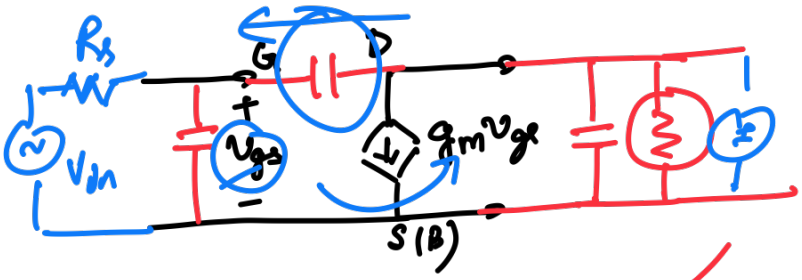
$$= f(\alpha V_{dd} + \beta V_{in})$$

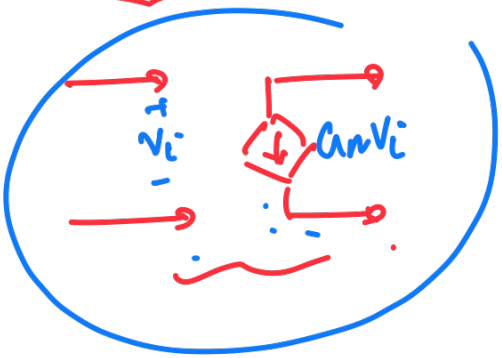


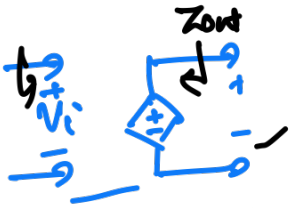
$I_{DS} = f(V_G, V_D, V_S, V_B, T)$

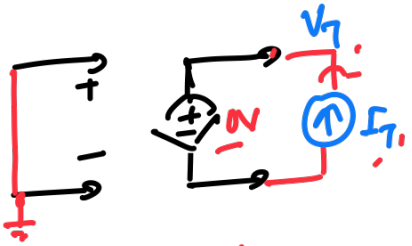










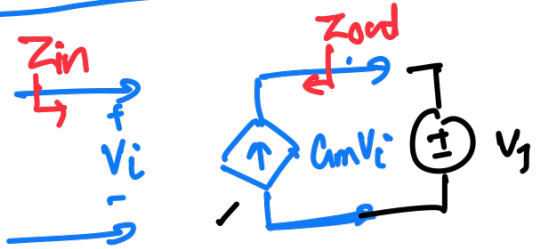


VCCS : Voltage Controlled Voltage Source.

i/p imp : ∞

$Z_{out} = 0$

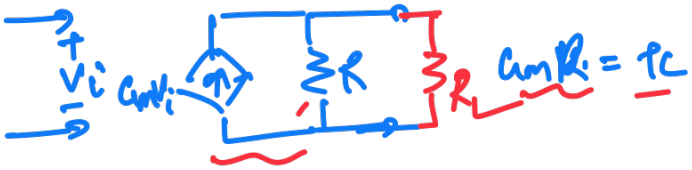
$\frac{V_T}{I_T} = Z_{out} = 0$

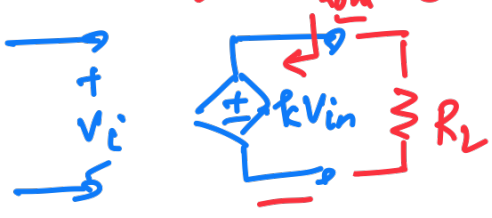


$Z_{in} = \infty$

$Z_{out} = \infty$

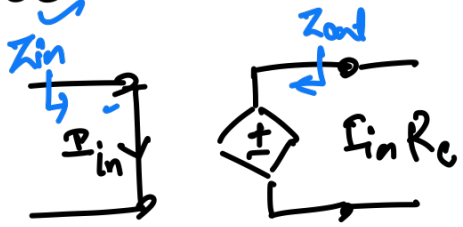
VCCS : Voltage Controlled Current Source.





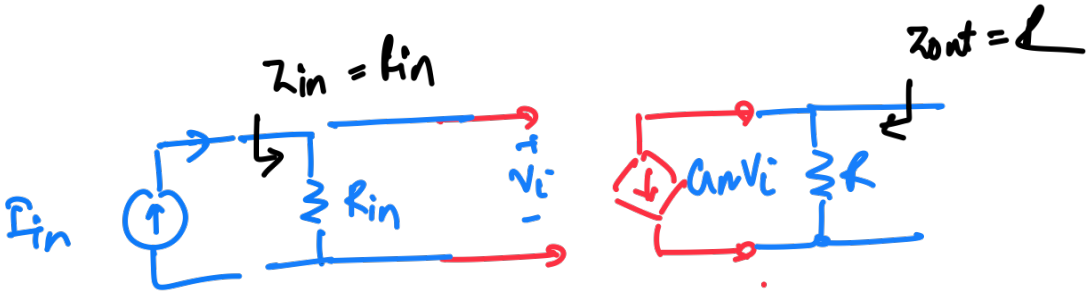
$A_m (R \| R_L) = T_C$

CCVS: Current-Controlled Voltage Source



$Z_{in} = 0$

$Z_{out} = 0$

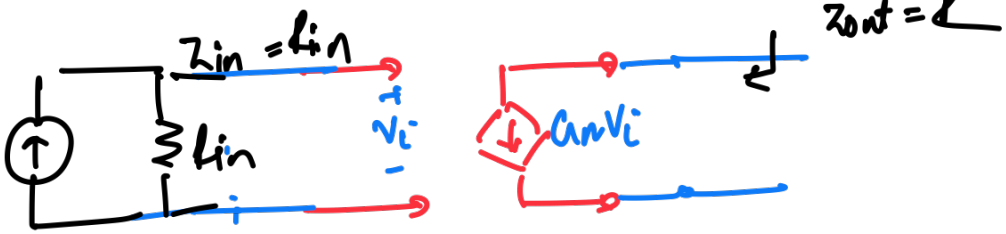


CCCS

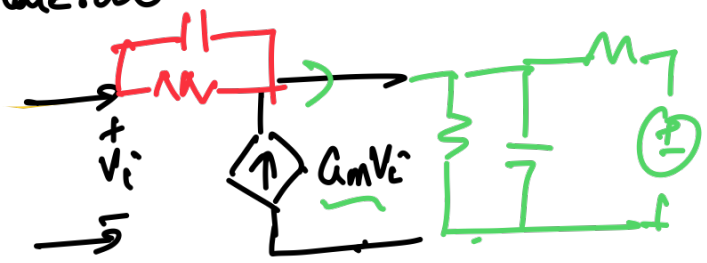


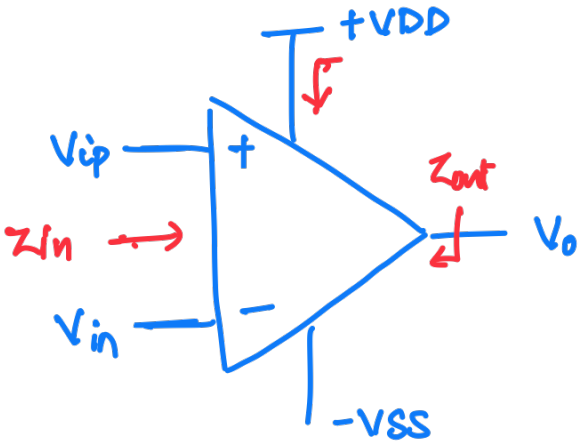
$Z_{in} = 0$

$Z_{out} = \infty$



- Unilateral





"Opamp."

$$V_o = A(V_{ip} - V_{in})$$

$Z_{in} = \infty$, $Z_{out} = 0$